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MILITARY SPECIFICATION

PROPELLANT, PRIMING FLUID, ALCM ENGINE, GRADE PF-1

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of high density hydrocarbon fuel.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Military

MIL-I-27686	Inhibitor, Icing, Fuel System

STANDARDS

Federal

FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-290	Packaging of Petroleum and Related Products

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.)

AMSC N/A FSC 9135

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government documents, form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AFR 71-4 Preparing Hazardous Materials for Military Air Shipments

IATA International Air Transport Association

ICAO International Civil Aviation Organizations

International Maritime Dangerous Goods Code

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington DC 20402).

2.1.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society for Testing and Materials Standards

ASTM D 93	Test Methods for Flash Point by Pensky-Martens Closed Tester
ASTM D 156	Test Methods for Saybolt Color of Petroleum Products (Saybolt Chromometer Method)
ASTM D 240	Standard Test Method For Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter
ASTM D 381	Standard Test Method for Existent Gum in Fuels by Jet Evaporation
ASTM D 445	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)
ASTM D 1298	Standard Test Method for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method
ASTM D 1655	Specification of Aviation Turbine Fuels
ASTM D 2276	Standard Test Methods for Particulate Contaminant in Aviation Turbine Fuels
ASTM D 2382	Standard Test Method for Heat of Combustion of Hydrocarbon Fuels by Bomb Calorimeter High-Precision Method)
ASTM D 2386	Standard Test Method for Freezing Point of Aviation Fuels
ASTM D 3241	Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels (JFTOT Procedures)
ASTM D 3828	Standard Test Method For Flash Point By Setaflash Closed Tester
ASTM D 4057	Standard Practice for Manual Sampling of Petroleum and Petroleum Products

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ASTM E 29 Recommended Practice of Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

Department of Transportations

CFR Title 49, Department of Transportation Rules an Regulations of the Parts 100-199 Transportation of Explosives and Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated details specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Except as otherwise specified herein, the priming fluid shall consist completely of a mixture of exo-tetrahydrodi(cyclopentadiene) and methylcyclohexane.

3.1.1 Reclaimed materials. There is no exclusion to the use of recovered materials. Reclaimed materials shall be required to the maximum extent possible.

3.2 Chemical and physical requirements. The chemical and physical requirements of the priming fluid shall conform to those listed in table I.

3.2.1 Additives. The additives listed shall be used in combination in amounts not to exceed those specified. The type and amount of each additive used shall be reported.

3.2.2 Antioxidants. The following active inhibitors shall be blended separately or in combination into the priming fluid in total concentration of 90 parts per million (minimum) to 110 parts per million (maximum), by weight, not including weight of solvent, in order to prevent the formation of polymeric oxidation products:

a. 2,6-di-tert-butyl-4-methylphenol

b. 6-tert-butyl-2,4-dimethylphenol

c. 2,6-di-tert-butylphenol.

3.2.3 Fuel system icing inhibitor. The fuel system icing inhibitor shall conform to MIL-I-27686.

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TABLE I. Chemical and physical requirements and test methods.

Property	Priming Fluid Min	Priming Fluid Max	ASTM Standards
Color, Saybolt	+25		D 156
Chemical composition, wt, percent			
methylcyclohexane	10.0	15.0	1/
exo-tetrahydrodi (cyclopentadiene)	83.5	90.0	
other hydrocarbons		1.5	
Flash point, °C (°F)	16 (60)	27 (80)	D 93, D 3828
Specific gravity, (60/60F)	0.900	0.930	D 1298
Freezing point, °C (°F)		-54 (-65)	D 2386
Viscosity, centistokes at -54°C		40	D 445
Net heat of combustion MJ/kg (Btu/lb)	42.1 (18,100)		D 240, D 2382
Thermal stability change in			D 3241 2/
pressure drop, mm Hg		10	
heater tube deposit visual rating		< Code 3	
Existent gum, mg/100 mL		5	D 381
Particulate matter, mg/liter		1.0	D 2276
Fuel system icing inhibitor, volume percent	0.10	0.15	3/

1/ Test shall be performed using Appendix A.

2/ See 4.5.3 for D 3241 test conditions and test limits.

3/ Test shall be performed using method 5327 or 5342 of FED-STD-791.

3.3 Performance. The priming fluid shall satisfy the performance requirements as specified in the tests of section 4, herein.

3.4 Workmanship. The priming fluid shall be visually free from undissolved water, sediment or suspended matter and shall be clean and bright at the ambient temperature or at 21°C (70°F), whichever is higher.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements,however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Material inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 Material inspection. Inspection shall be performed in accordance with method 9601 of FED-STD-791.

4.4 Quality conformance inspection. For acceptance purposes, individual lots shall be examined as specified herein and subject to tests to determine compliance with all requirements in section 3.

4.4.1 Inspection lot

4.4.1.1 Bulk lot. A bulk lot shall consist of an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container.

4.4.1.2 Packaged lot. A packaged lot shall consist of an indefinite number of 55-gallon drums or smaller unit package of identical size and shape offered for acceptance and filled from the isolated tank containing a homogeneous mixture of material.

4.4.2 Sampling

4.4.2.1 Sampling for verification of product quality. Each bulk or packaged lot of material shall be sampled for verification of product quality in accordance with ASTM D 4057, except where individual test procedures contains specific sampling instructions.

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4.4.2.2 Sampling for examination of filled containers for delivery. A random sample of filled containers shall be selected. The samples shall be examined in accordance with 4.5.1.3.

4.5 Inspection methods

4.5.1 Examination

4.5.1.1 Examination of product. Samples selected in accordance with 4.4.2.1 shall be visually examined for compliance with 3.4.

4.5.1.2 Examination of empty containers. Prior to filling, each empty unit container shall be visually inspected for cleanliness and suitability.

4.5.1.3 Examination of filled containers. Samples taken as specified in 4.4.2.2 shall be examined for conformance to MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and markings. Any container having one or more defects under the required fill shall be rejected.

4.5.2 Chemical and physical tests. Tests to determine conformance to chemical and physical requirements shall be conducted in accordance with table I. The priming fluid shall pass all tests listed in table I. No additional testing shall be required. Requirements contained herein are not subject to corrections for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances may be averaged. For rounding off of significant figures, ASTM E 29 shall apply to all tests required by this specification.

4.5.3 Thermal stability test. The thermal stability test shall be conducted using ASTM D 3241. The heater tube shall be rated visually (see Appendix B).

4.5.3.1 Test conditions

a. Heater tube temperature at maximum point: 260°C (500°F),

b. System pressure: 3.45 megaPascal (500 pounds per square inch gage).

c. Flow rate: 3.0 milliliter/minute.

d. Test duration: 150 minutes.

4.5.3.2 Test results. The sample is acceptable if all the following criteria are met:

a. The maximum visual rating of the heater tube deposits is less than a code 3 (Appendix B, 30.6).

b. The visual rating of the heater tube shows neither peacock type deposit (code P) nor abnormal type deposits (Code A) (Appendix B, 30.6.3.1 and 30.6.3.2).

c. The maximum differential pressure across the test filter does not exceed 10 millimeters of mercury.

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4.5.3.3 Reported data. The following data shall be reported;

a. Differential pressure in millimeters of mercury at 150 minutes or time to differential pressure of 10 millimeters of mercury, whichever comes first.

b. Heater tube deposit code rating at the end of the test.

4.6 Test report. Test data required by 4.5 shall be reported in the same order as listed in Table I. The Inspection Data on Aviation Turbine Fuels form published in ASTM D 1655 shall be used as a guide. Also, the type and amount of additives used shall be reported (6.3.1).

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with MIL-STD-290. All containers shall be marked with the actual flash point in degrees F of the priming fluid contained therein.

5.2 Transportation. The transportation of PF-1 fuel shall be in accordance with Code of Federal Regulations 49 (49 CFR), Parts 100-199; AFR 71-4, Preparing Hazardous Materials for Military Air Shipments; International Maritime Dangerous Goods Code; International Air Transport Association (IATA); and the International Civil Aviation Organizations (ICAO), as applicable for mode of transportation and destination.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The priming fluid covered by this specification is intended for use in the ALCM F107-WR-101 engine.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Data requirements. The following Data Item Description (DID) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475.1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
4.5.3.3	DI-T-3733	Quality Conformance Test Reports (Fuels)

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.3.1 Type and amount of additive (3.2.1).

6.4 Key words

ALCM engine

Inhibitor

Priming fluid

Propellant

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 68

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APPENDIX A

METHOD OF TEST FOR ANALYSIS OF GRADE PF-1 PROPELLANT BY GAS CHROMATOGRAPHY

10. Scope. This method is intended for the analysis of PF-1 Propellant. The method provides for the quantitative determination of the two major constituents of PF-1 and for the characterization of each constituent.

20. Summary of methods. The sample is introduced into a gas chromatographic column and separated into its individual components. Conditions are selected to give complete resolution of the two major blending stocks. The detector signal is monitored continuously and the areas of individual peaks are used for computing the quantities of the components. Response factors are applied to give the fuel composition in true weight percent. The detailed composition of each major component is computed from the individual peak areas in the subgroup.

30. Apparatus

30.1 Chromatograph. Any conventional gas chromatograph, having at least the following features, is acceptable.

30.1.1 Detector. Either a thermal conductivity (TCD) or flame ionization (FID) detector may be used. The detector shall be operated in such a manner that its linear dynamic range is not exceeded, and its linearity should be checked periodically. The detector shall be capable of continuous operation at 325°C or higher, and must be connected to the column in a manner that eliminates cold spots.

30.1.2 Temperature programmer. The instrument must be capable of reproducible temperature programming over the range of 50° to 300°C. The programming rate must be sufficiently reproducible that individual retention times do not vary by more than two percent. Any temperature program, including multilinear modes and intermediate periods of isothermal operation, is acceptable, provided that the column resolution, as defined in section 50.2, is not compromised.

Note 1. If capillary-type columns are used for the analysis, it may not be necessary to program the oven above 200°C.

30.1.3 Sample inlet system. Either flash vaporization or on-column injection may be used. For flash vaporization, the sample inlet port must be maintained between 300° and 350°C, and well-conditioned septa must be used. If on-column injection is employed, provision must be made for programming the temperature of the full length of the column. A sample inlet splitter may be used in conjunction with capillary columns, if care is taken to ensure that a representative sample is delivered to the column.

Note 2. New septa should be allowed to condition overnight after being installed in order to minimize spurious peaks, and several blank runs should be made to purge the column of materials bleeding from the septum. Overnight conditioning can be eliminated by the expedient of storing a supply of septa in the column oven. However, at least one blank run should still be made following each septum change.

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30.1.4 Recorder. Any potentiometric recorder compatible with the chromatograph may be used. The full-scale response time should be two seconds or less, and the gain and damping controls should be adjusted to provide optimum response and minimum noise.

30.1.5 Column. Either packed or open-tubular (capillary-type) columns may be used, provided the minimum resolution, specified in section 50.2, is maintained. The column should be designed for maximum thermal stability and minimum bleed, since the temperature program is likely to run as high as 275°C for packed column and 200°C for capillary columns. Dual-column compensation is required if column bleed results in excessive baseline drift at high temperatures.

30.1.6 Integrator. An electric integrator is required for highest accuracy and precision. Data acquisition and reduction by an on-line computer, programmed for gas chromatography, is convenient and may be used if such capability is available.

30.1.7 Microsyringe. A microsyringe, capable of delivering sample volumes from 0.5 to 5.0 microliters, is required.

Note 3. The widely used 0 - 10 microliter syringe, Hamilton Model 701N or its equivalent, is recommended. These syringes are routinely available from most instrument manufacturers and chromatography supply houses.

40. Reagents and materials

40.1 Stationary-phase. The stationary phase should be one of the familiar non- polar silicones. The following silicones have been used successfully:

Silicone SE-30

Silicone UCW-982

Silicone OV-101

Note 4. UCW-982 is recommended because it exhibits less column bleed at high temperatures than the other silicones. However, any of the listed phases are satisfactory, provided baseline drift at high temperature is minimized by conditioning or by dual column compensation.

40.2 Carrier gas. Either helium or nitrogen is suitable, although nitrogen is known to give somewhat higher column efficiencies at lower flow rates, and is more economical. Helium must be used with thermal conductivity detectors. The flow must be maintained constant to within one percent throughout the entire range of operating temperatures by means of constant flow controllers.

40.3 Hydrogen. High purity, for flame ionization detectors.

40.4 Air. Dry, filtered, and oil-free, for flame ionization detectors.

40.5 exo-Tetrahydrodi(cyclopentadiene). (99 percent purity), hereafter referred to as C-10.

Note 5. High purity samples of C-10 may be obtained by submitting a written request to WRDC/POSF, Wright-Patterson AFB OH 45433-6563.

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40.6 Methylcyclohexane. (99 percent purity) hereinafter referred to as MCH.

40.7 n-Pentane, n-hexane, or carbon disulfide. Chromatographic purity.

40.8 Qualitative calibration mixture. A series of known hydrocarbons is used for calculating column resolution. Prepare a mixture containing about one percent (by volume) each, in pentane, hexane, or carbon disulfide of at least eight normal alkanes and six cycloalkane derivatives selected from highly pure (minimum 99 percent) samples which have boiling points in the 70°C-250°C range. The mixture must contain MCH, n-dodecane, and n-tetradecane.

40.9 Quantitative calibration mixture. Prepare several calibration mixtures by carefully weighing quantities of MCH and C-10 into sample vials or other suitable closed containers. Handling and weighing should be performed carefully so that the weight of each component is accurately known to within one percent (relative).

Note 6. The calibration mixtures shall be of varying compositions in order to confirm the linearity of the detector and the validity of the calibration; response factors should be constant and independent of the actual compositions. The compositions should approximate the range defined in the PF-1 specification but it is helpful to prepare the samples so that these limits are occasionally exceeded.

Note 7. Weighing may be done by difference into any convenient containers, provided care is taken to introduce the materials quickly and thereby avoid loss by evaporation. Since MCH has appreciable vapor pressure under ambient conditions, it should be weighed last. The containers should be leak-tight, and stored in a cold box (preferably a freezer compartment.) Properly prepared and stored samples are stable for several months.

50. Preparation of apparatus

50.1 Column. Any conventional method of conditioning and installing the column is satisfactory, provided the system is free of gas leaks and the column resolution meets the minimum requirement specified in section 50.2

50.2 Column resolution. Run a sample of the qualitative calibration mixture and measure the retention times and widths of the n-dodecane and n-tetradecane peaks as shown in figure A1. Resolution is defined as the difference in retention times for n-C_{12} and n-C_{14} divided by their average width:

$$R = \frac{2(RT_{14} - RT_{12})}{W_{14} + W_{12}}$$

Resolution determined in this way must be at least 5.0 and can be optimized by varying the temperature program and the carrier flow rate. If resolution cannot be increased to at least 5.0 by optimizing these parameters, it is necessary to use a different column.

50.3 Chromatograph. Place in service according to the manufacturer's specifications paying particular attention to flow rates through the detector.

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60. Procedure

60.1 Establish a suitable sample volume by making several trial runs with the qualitative calibration mixture. For FID instruments, the volume may be between 0.5 and 1.0 microliter; for TCD instruments, volumes from 1.0 to 5.0 microliters have been found satisfactory.

Note 8. In establishing the optimum temperature program, it is necessary to begin the run at a low enough temperature to allow complete elution of the solvent before the appearance of the MCH peak. If the area is being monitored by an electronic integrator or computer, a time delay should be entered to inhibit integration of the solvent peak(s).

60.2 Dilute an aliquot of each quantitative calibration mixture at least 1:10 (preferably 1:20) with either pentane, hexane, or carbon disulfide. Mix thoroughly and inject a measured volume into the chromatograph, simultaneously activating the temperature program, the integrator, and the recorder.

60.3 Integrate the area under the chromatogram continuously until the final peak has been eluted.

60.4 Compute a response factor for MCH as described in section 70. Repeat the procedure at least three times for each calibration sample and find the mean value for each response factor. Then dilute the PF-1 acceptance sample and run in the same way as the calibration standards, and determine its composition as described in section 70.

Note 9. For MCH and C-10, extra peaks which are within plus or minus ten percent of the retention time for MCH or C-10, respectively, shall be considered part of the MCH or C-10 peak areas. If commercial grade MCH was used in blending the PF-1 acceptance sample, appreciable amounts of heptane and toluene may appear as resolved peaks near the MCH peak. These peak areas should be added, uncorrected, to the area of the MCH peak. Assuming unit relative response factors for these materials does not seriously affect the accuracy of the method.

70. Calculations

70.1 C-10 is used in each case as an internal standard, and is considered to have a response factor of 1.000. From the analysis of each calibration standard group together the peak areas corresponding to the MCH and calculate the relative response for MCH by

$$K_{MCH} = \frac{wt(MCH)}{wt(C\text{-}10)} \times \frac{area(C\text{-}10)}{area\ (MCH)}$$

70.2 For each PF-1 acceptance sample, apply the calculated MCH response factor to each MCH peak area, and sum the products to obtain a corrected MCH area. In the following equation, all hydrocarbons other than those contained in the MCH and C-10

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subgroups are considered impurities. A response factor of 1.000 is used for the impurities. Calculate the weight percent of each PF-1 component as follows:

$$\%MCH = \frac{K_{MCH} A_{MCH}}{K_{MCH} A_{MCH} + A_{C-10} + A_{Impurities}} X\ 100$$

$$\%C-10 = \frac{A_{C-10}}{K_{MCH} A_{MCH} + A_{C-10} + A_{Impurities}} X\ 100$$

$$\%\ \text{Other hydrocarbons} = 100 - (\%MCH + \%\ C-10)$$

80. Precision. Detailed precision studies have not been performed for this method. However, limited results indicate that the method is capable of the following precision.

80.1 Repeatability. Replicate analyses, performed under identical conditions in the same laboratory, should yield results that do not vary more than one percent (absolute) from the mean value for each component.

80.2 Reproducibility. When the same sample is analyzed in different laboratories according to this method, the results for any component should not differ by more than 1.5 percent (absolute.)

80.3 Accuracy. Back-analysis of a calibration standard should give results that agree with the known values to within 0.5 percent (absolute.)

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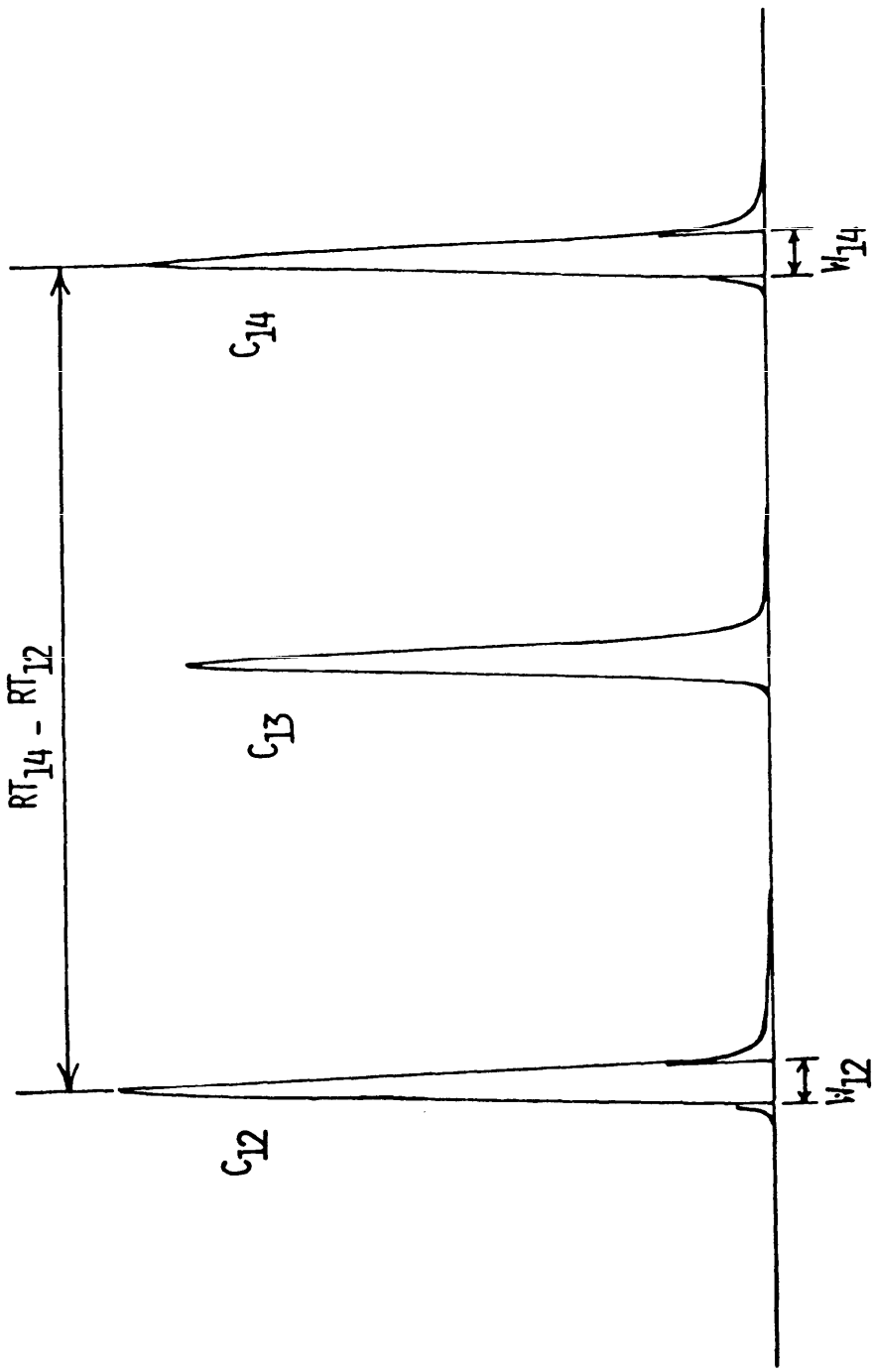


FIGURE A1. Calculation of column resolution.

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APPENDIX B

HEATER TUBE DEPOSIT RATING

10. SCOPE

10.1 Scope. This appendix provides instructions for conducting and reading the heater tube deposit. This appendix is a mandatory part of the specification.

20. REFERENCED DOCUMENTS. This section is not applicable to this appendix.

30. VISUAL TEST METHOD

30.1. Snap the upper end of the heater tube into the clamp of the adapter for the heater tube.

30.2 Push the heater tube against the stop of the adapter tube.

30.3 Slide the adapter with the heater tube over the guide rod into the tuberator equipped with a magnifying glass assembly.

30.4 Insert the ASTM color standard into the tuberator.

30.5 Rotate the adapter and position the heater tube so that the side with the maximum deposit is visible.

30.6 Within 30 minutes after completion of the test, visually examine the heater tube in a tuberator. The entire portion of the test section between the bottom shoulder and the top shoulder of the heater tube test section shall be carefully examined using a magnifying glass in conjunction with the tuberator for any signs of discoloration, scratches, or other visually identified defects. When an area of the tube corresponds visually to an ASTM color standard, that color standard code number shall be recorded. If the area being rated has a color between two adjacent color standards, it shall be rated as the lighter (that is lower number) color standard. (NOTE: It is important that all light bulbs in the tuberator are functioning as a change in light intensity can shift the rating significantly.) (NOTE: The person rating the tube should have normal ability to distinguish between colors; i.e., he should not be color blind.)

30.6.1 In rating the heater tube, the darkest deposits govern and the code number representative of the darkest section, rather than the average deposit, shall be reported.

30.6.2 If a spot or streak is found on the heater tube, it shall be carefully examined under various lighting conditions using a magnifying glass to determine if it is a deposit, a scratch, or tube defect (note that the tube defects should have been found during the pretest inspection of the tube). If the spot or streak is determined to be a scratch or tube defect, it shall be disregarded.

If the spot or streak is a deposit, it shall be rated against the ASTM color standards, if larger in area than about 0.004 square inch (0.025 sq cm); i.e., approximately 1/16 inch x 1/16 inch (1.5 cm x 1.5 cm) square or an equivalent area. However, a streak deposit shall be ignored if less than 1/32 inch (0.8 mm) wide, regardless of length. Note that the tube section is 1/8 inch (about 3 mm) in diameter; thus a 1/16 inch

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(1.5 mm) wide spot is half the diameter of the tube test section and 1/32 inch (0.8 mm) wide streak is one fourth the diameter of the tube test section.

30.6.3 If the heater tube has deposits which do not match the Color Standards, the following criteria shall be used:

30.6.3.1 If the deposit has peacock (rainbow) colors, rate this as Code P (P for peacock). If some portion of the deposit does match the Color Standard, it shall be rated.

30.6.3.2 Deposits having abnormal colors (for example, blue or gray) shall have rating of code A (A for abnormal color) assigned.

30.6.3.3 When reporting the overall tube rating, record the rating of the maximum deposit which matches the Color Standards plus P or A if the tube contains deposits which do not match the Color Standards. If the tube contains only P or A deposits, just report the appropriate letter(s); do not try to assign a numerical rating to a P or A deposit. Examples of how the rating procedure is to be used are given below:

Example 1: The darkest deposits on the heater tube match color standard 3. Also present are peacock colors. Thus, the overall tube rating to be reported in 3P.

Example 2: The heater tube has maximum deposits falling between Color Standards 2 and 3 and has no peacock or abnormal colors. The total tube rating is 2.

Example 3: The heater tube matches Color Standard 1 except for an abnormal deposit which does not match the ASTM Color Standards. The overall tube rating to be reported is 1A.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-87173A(USAF)	Propellant, Priming Fluid, ALCM Engine, Grade PF-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:_____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)*–Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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